5.7 Using Congruent Triangles

Name _____ Period ____ Date _____

Geometry 5.7 · Student sheet

I can…

- use congruent triangles to find a length I can't measure directly.
- use CPCTC in a proof.
- plan a proof that uses congruent triangles.

Vocabulary · fill in the missing word

CPCTC — _____ parts of congruent triangles are congruent

Part 1 · Discovery: measure what you can't reach

Mark the river, walk the steps, and measure.

Distance from B to C	Distance from C to D	DE	Distance across (AB)

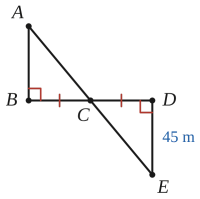
Why does DE equal AB?

Part 2 · Notes and examples (fill in each note, then work the example)

CPCTC: once two triangles are proven congruent, all of their _____ parts are congruent.

Example 1

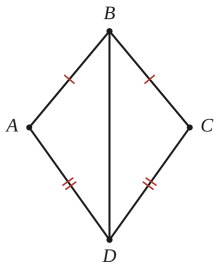
In the diagram, $\triangle ABC \cong \triangle EDC$ (ASA: right angles at B and D, $\overline{BC} \cong \overline{DC}$, vertical angles at C). DE = 45 m. How wide is the river (AB)?



Plan: to prove two parts congruent, find triangles that contain them, prove the triangles congruent _____, then use CPCTC.

Example 2

Given: $\overline{AB} \cong \overline{CB}$ and $\overline{AD} \cong \overline{CD}$. Prove: $\angle A \cong \angle C$.



Example 3

Given: M is the midpoint of $\overline{AB}$ and of $\overline{CD}$. Prove: $\overline{AC} \parallel \overline{BD}$.

Example 4

$\triangle PQR \cong \triangle STU$. $PQ = 2x + 1$ and $ST = 15$. $m\angle R = 40°$. Find x and $m\angle U$.

Part 3 · Partner practice

Partner 1

Given: $\angle B \cong \angle D$ and $\overline{AC}$ bisects $\angle BAD$. Prove: $\overline{BC} \cong \overline{DC}$.

Partner 2

$\triangle JKL \cong \triangle MNO$ by SSS. $m\angle K = 72°$. Find $m\angle N$.